# Chan Young Joe

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## Current Position

**Researcher, Data Science Team** — Jan. 2026 – Present
*Research Institute of Environmental Health and Safety, EH R&C Co.* — *Incheon, Korea*

## Education

**University of Seoul** — Seoul, Korea
*M.S. in Applied Chemistry (Computational Chemistry), advisor Prof. Rakwoo Chang* — *Sep. 2024 – Feb. 2026*

**University of Seoul** — Seoul, Korea
*B.S. in Applied Chemistry; B.E. in Artificial Intelligence (double major)* — *Mar. 2021 – Aug. 2024*

**National Institute for Lifelong Education (NILE), Korea University** — Seoul, Korea
*B.S. in Mathematics* — *Jan. 2025 – Dec. 2026 (expected)*

## Research Experience

**Researcher, Computational Molecular Modeling Laboratory** — Jun. 2021 – Feb. 2026
*University of Seoul (Prof. Rakwoo Chang)* — *Seoul, Korea*

- Large language models (in-context learning, fine-tuning) for molecular property prediction
- Molecular dynamics of membranes and enzymes; hybrid multigrained sampling

**Visiting Researcher, Department of Bioengineering** — Jun. 2024 – Aug. 2024
*Lehigh University (Prof. Wonpil Im)* — *Bethlehem, PA, USA*

## Selected Publications

Lee, J.-H.*; **Joe**, **C. Y.** A flux-based bioaccumulation model for growing organisms: integrating advective–diffusive uptake at exchange organs into dynamic energy budget theory. *Environ. Toxicol. Chem.* **2026**, vgag253, DOI: 10.1093/etojnl/vgag253.

**Joe**, **C. Y.**; Song, K.*; Chang, R.* Evaluating in-context learning in large language models for molecular property regression. *J. Comput. Chem.* **2026**, *47*, e70308, DOI: 10.1002/jcc.70308.

## Open-Source Software

**PFAS–rice uptake model** | *Python, HYDRUS-1D, Streamlit* — github.com/ccy5123/pfas-rice-model

- Mechanistic four-compartment (root–stem–leaf–grain) uptake model for anionic PFAS in paddy rice, covering 12 calibrated PFCA/PFSA congeners
- Coupled to HYDRUS-1D soil transport; parameterisable from chemical structure; interactive web application

**sci-adk** | *Python, Claude Code* — github.com/ccy5123/sci-adk

- Verification kit for agent-assisted research: AI agents propose, and a deterministic engine judges each claim against decision rules frozen before the experiment
- Append-only evidence record with headless re-verification by third parties and tamper-evident provenance

## Selected Presentations

**Joe**, **C. Y.**; Lee, J.-H.* Integrated DEB-TK model for fish bioaccumulation assessment of neutral organic chemicals: theoretical formulation and empirical validation. Presented at the 2026 Spring General Meeting of the Korean Society of Environmental Health, Jeju, Korea, May 2026; poster P-EaRA-40. Best Poster Award.

**Joe**, **C. Y.**; Song, K.*; Chang, R.* Evaluating in-context learning in large language models for molecular property regression. Presented at Pacifichem 2025, Honolulu, HI, Dec. 15–20, 2025; poster PHY006.

*Corresponding author

## Honors

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| Best Poster Award, 2026 Spring General Meeting of the Korean Society of Environmental Health | 2026 |
| Outstanding Paper Award, President of the University of Seoul (*J. Comput. Chem.* paper) | 2026 |
| Best Poster Award, System Chemistry Global Leading Research Center Winter Symposium | 2025 |
| Best Poster Award, Fall Meeting of the Korean Society of Environmental Health and Toxicology | 2024 |
| President's Award, graduation with honors (cum laude), University of Seoul | 2024 |